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Algorithm

Main Program

1. Start.
2. Display the menu:
    - Find the longest word in a sentence.
    - Check if one string is a rotation of another.
    - Exit.
3. Read the user's choice.
4. Use `switch` to call the corresponding function.
5. Repeat the menu until the user selects Exit.
6. Stop.

Function 1: Find Longest Word

1. Read a sentence.
2. Scan the sentence word by word.
3. Calculate the length of each word.
4. Compare each word's length with the maximum length found so far.
5. Store the longest word.
6. Display the longest word and its length.

Function 2: Check String Rotation

Two strings are rotations if one can be obtained by moving characters from the beginning of the other to the end.

Example: `ABCD` → `CDAB`

1. Read two strings.
2. Compare their lengths.
3. If lengths are different, they cannot be rotations.
4. Concatenate the first string with itself.
5. Search for the second string inside the concatenated string using `strstr()`.
6. If found, the strings are rotations; otherwise, they are not.

C Program

```
#include <stdio.h>

#include <string.h>

#include <ctype.h>
```

```
#define MAX 200

void findLongestWord(void);
void checkRotation(void);

int main(void)
{
    int choice;

    do
    {
        printf("\n===== HDOC Day-43 =====\n");
        printf("1. Find the longest word in a sentence\n");
        printf("2. Check if one string is a rotation of another\n");
        printf("3. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();  // remove newline left by scanf

        switch (choice)
        {
            case 1:
                findLongestWord();
                break;
```

```
            case 2:
                checkRotation();
                break;

            case 3:
                printf("Exiting program...\n");
                break;

            default:
                printf("Invalid choice. Please enter 1, 2, or 3.\n");
        }

    } while (choice != 3);

    return 0;
}

/* Function to find the longest word */
void findLongestWord(void)
{
    char sentence[MAX];
    char longest[MAX];
    char current[MAX];
```

```
int i = 0, j = 0;
int maxLength = 0;

printf("Enter a sentence: ");
fgets(sentence, sizeof(sentence), stdin);

while (1)
{
   /* Store characters belonging to a word */
   if (isalpha((unsigned char)sentence[i]))
   {
      current[j] = sentence[i];
      j++;
   }
   else
   {
      /* End of a word */
      if (j > 0)
      {
         current[j] = '\0';

         if (j > maxLength)
         {
            maxLength = j;
            strcpy(longest, current);
```

```
            }


            j = 0;
        }


        /* End of sentence */
        if (sentence[i] == '\0')
            break;
    }


    i++;
  }


  if (maxLength > 0)
  {
    printf("Longest word: %s\n", longest);
    printf("Length: %d\n", maxLength);
  }
  else
  {       printf("No word found in the sentence.\n");
  }
}
/* Function to check whether two strings are rotations */
void checkRotation(void)
{
```

```
    char str1[MAX];
    char str2[MAX];
    char combined[2 * MAX];
    printf("Enter the first string: ");
    fgets(str1, sizeof(str1), stdin);
    printf("Enter the second string: ");
    fgets(str2, sizeof(str2), stdin);
    /* Remove newline characters */
    str1[strcspn(str1, "\n")] = '\0';
    str2[strcspn(str2, "\n")] = '\0';

    if (strlen(str1) != strlen(str2))
    {
        printf("The strings are NOT rotations of each other.\n");
        return;
    }
    strcpy(combined, str1);
    strcat(combined, str1);
    if (strstr(combined, str2) != NULL)
        printf("The strings are rotations of each other.\n");
    else
        printf("The strings are NOT rotations of each other.\n");
}
```

Test Case Table

Test	Operation	Input	Expected Output
1	Longest word	`C programming is interesting`	`programming`
2	Longest word	`I love computers`	`computers`
3	Longest word	`Hello world`	`Hello` (first longest)
4	Rotation	`ABCD`, `CDAB`	Rotation
5	Rotation	`ABCD`, `DABC`	Rotation
6	Rotation	`ABCD`, `ACBD`	Not rotation
7	Rotation	`HELLO`, `LOHEL`	Rotation
8	Rotation	`ABC`, `ABCD`	Not rotation
9	Menu	Choice `5`	Invalid choice
10	Menu	Choice `3`	Exit

Sample Execution

```
[aaryanrajput@aaryans-macbook ~ % clang day43.c -o day43.c
[aaryanrajput@aaryans-macbook ~ % ./day43.c

===== HDOC Day-43 =====
1. Find the longest word in a sentence
2. Check if one string is a rotation of another
3. Exit
Enter your choice: 1
Enter a sentence: hello my name is aaryan
Longest word: aaryan
Length: 6

===== HDOC Day-43 =====
1. Find the longest word in a sentence
2. Check if one string is a rotation of another
3. Exit
Enter your choice: 2
Enter the first string: abcd
Enter the second string: cdab
The strings are NOT rotations of each other.

===== HDOC Day-43 =====
1. Find the longest word in a sentence
2. Check if one string is a rotation of another
3. Exit
Enter your choice: 3
Exiting program...
aaryanrajput@aaryans-macbook ~ %
```